

Proof of The Chain Rule

Benjamin Abshire[†]
Kent State University

January 15, 2026

I Introduction

To prove the chain rule we must use Carathéodory's Theorem. The Theorem states that, for $f : I \rightarrow \mathbb{R}, c \in I$, f is differentiable at c if and only if there exists $\varphi : I \rightarrow \mathbb{R}$ that is continuous at c and satisfies $f(x) - f(c) = \varphi(x)(x - c) \forall x \in I$. Moreover, if either condition holds, then $\varphi(c) = f'(c)$.

II The Chain Rule

Let $I, J \subseteq \mathbb{R}$ be intervals. Also let $g : I \rightarrow \mathbb{R}, f : J \rightarrow \mathbb{R}, f(J) \subseteq I$ with $c \in J$. If f is differentiable at c and g is differentiable at $f(c)$, then $g \circ f$ is differentiable at c , with

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c) \tag{1}$$

Proof. Since there exists $f'(c)$, Carathéodory's Theorem implies that there exists a function φ on J such that φ is continuous at c and $f(x) - f(c) = \varphi(x)(x - c)$ (since f is differentiable) $\forall x \in J$ and where $\varphi(c) = f'(c)$. Also, there exists $g'(f(c))$ so there is a function ψ defined on J such that ψ is continuous at $f(c)$ and $g(y) - g(f(c)) = \psi(y)(y - f(c))$ (because g is differentiable) $\forall y \in I$, where $\psi(f(c)) = g'(f(c))$. Substituting $y = f(x)$ gives us

$$g(f(x) - g(f(c))) = \psi(f(x))(f(x) - f(c)) = \left[(\psi \circ f)(x) \cdot \varphi(x) \right] (x - c)$$

for all $x \in J$ such that $f(x) \in I$. Since $(\psi \circ f) \cdot \varphi$ is continuous at c (because ψ, f , and φ are all continuous) and its value at c is $g'(f(c)) \cdot f'(c)$, by Carathéodory's Theorem $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$. $\square$

[†]Definitions and example problems from MATH 42002: Analysis II instructed by Dr. Zachary Chase and *Introduction to Real Analysis* by Robert G. Bartle and Donald R. Sherbert